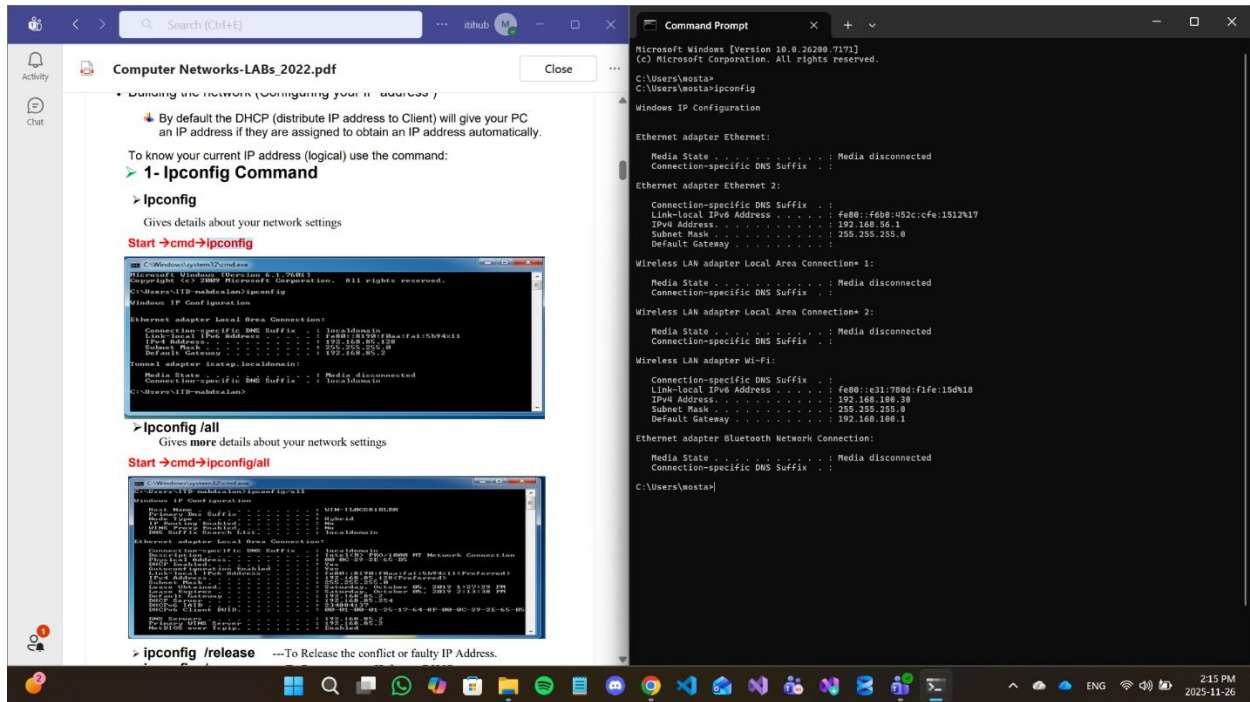
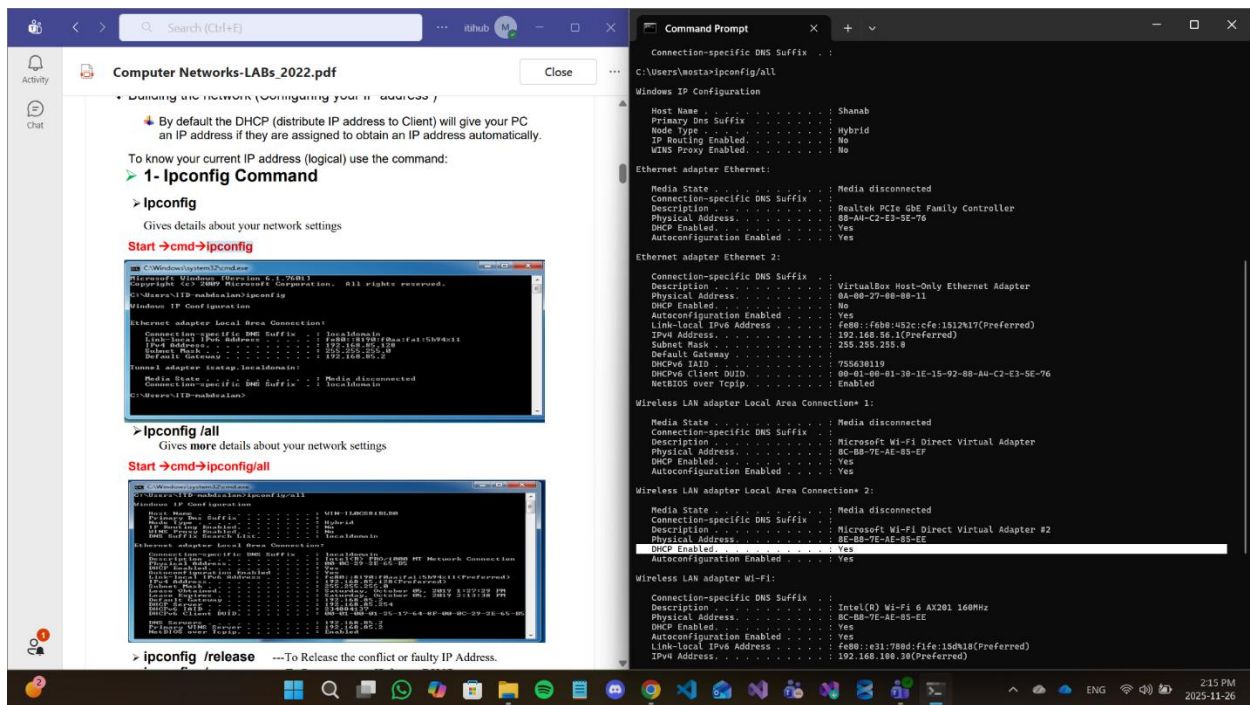




Search (Ctrl+E) rthub

No internet. Reconnect to keep the conversation going.

Computer Networks-LABs_2022.pdf Close

Activity Chat

Command Prompt

```
Lease Obtained. . . . . : Sunday, 23 November, 2025 5:26:38 PM
Lease Expires . . . . . : Thursday, 27 November, 2025 10:18:11 AM
Default Gateway . . . . : 192.168.100.1
DHCP Server . . . . . : 192.168.100.1
DHCPv6 I/D . . . . . : 168027234
DHCPv6 Client GUID . . . : 68-93-88-93-38-1E-15-92-88-A4-C2-E3-5E-76
DNS Servers . . . . . : 163.121.128.134
                        163.121.128.135
NetBIOS over Tcpip. . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . . : 
Description . . . . . : Bluetooth Device (Personal Area Network)
Physical Address. . . . . : 8C-8B-7E-AF-48-F2
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

C:\Users\mosta>ipconfig/release

Windows IP Configuration

No operation can be performed on Ethernet while it has its media disconnected.
No operation can be performed on Local Area Connection* 1 while it has its media disconnected.
No operation can be performed on Local Area Connection* 2 while it has its media disconnected.
No operation can be performed on Bluetooth Network Connection while it has its media disconnected.

Ethernet adapter Ethernet:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . . : 

Ethernet adapter Ethernet 2:

Connection-specific DNS Suffix . . : 
Link-local IPv6 Address . . . . . : fe80::f6b0:4b2c:cfe:1512%17
IPv6 Address. . . . . : 192.168.56.1
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 

Wireless LAN adapter Local Area Connection* 1:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . . : 

Wireless LAN adapter Local Area Connection* 2:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . . : 

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . . : 
Link-local IPv6 Address . . . . . : fe80::e31:78bd:f1fe:15d%18
Default Gateway . . . . . : 

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . . : 

C:\Users\mosta>
```

2:16 PM 2025-11-26

Search (Ctrl+E) rthub

Computer Networks-LABs_2022.pdf Close

Activity Chat

LABS

labs are proving the theoretical concepts that have been taught in lectures and enhance the technical skills for learners



**NETWORK
FUNDAMENTALS
LABS**

Prepared by : Mohamed Abesehly

Command Prompt

```
Wireless LAN adapter Local Area Connection* 2:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . . : 

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . . : 
Link-local IPv6 Address . . . . . : fe80::e31:78bd:f1fe:15d%18
Default Gateway . . . . . : 

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . . : 

C:\Users\mosta>ipconfig/renew

Windows IP Configuration

No operation can be performed on Ethernet while it has its media disconnected.
No operation can be performed on Local Area Connection* 1 while it has its media disconnected.
No operation can be performed on Local Area Connection* 2 while it has its media disconnected.
No operation can be performed on Bluetooth Network Connection while it has its media disconnected.

Ethernet adapter Ethernet:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . . : 

Ethernet adapter Ethernet 2:

Connection-specific DNS Suffix . . : 
Link-local IPv6 Address . . . . . : fe80::f6b0:4b2c:cfe:1512%17
IPv6 Address. . . . . : 192.168.56.1
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 

Wireless LAN adapter Local Area Connection* 1:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . . : 

Wireless LAN adapter Local Area Connection* 2:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . . : 

Wireless LAN adapter Wi-Fi:

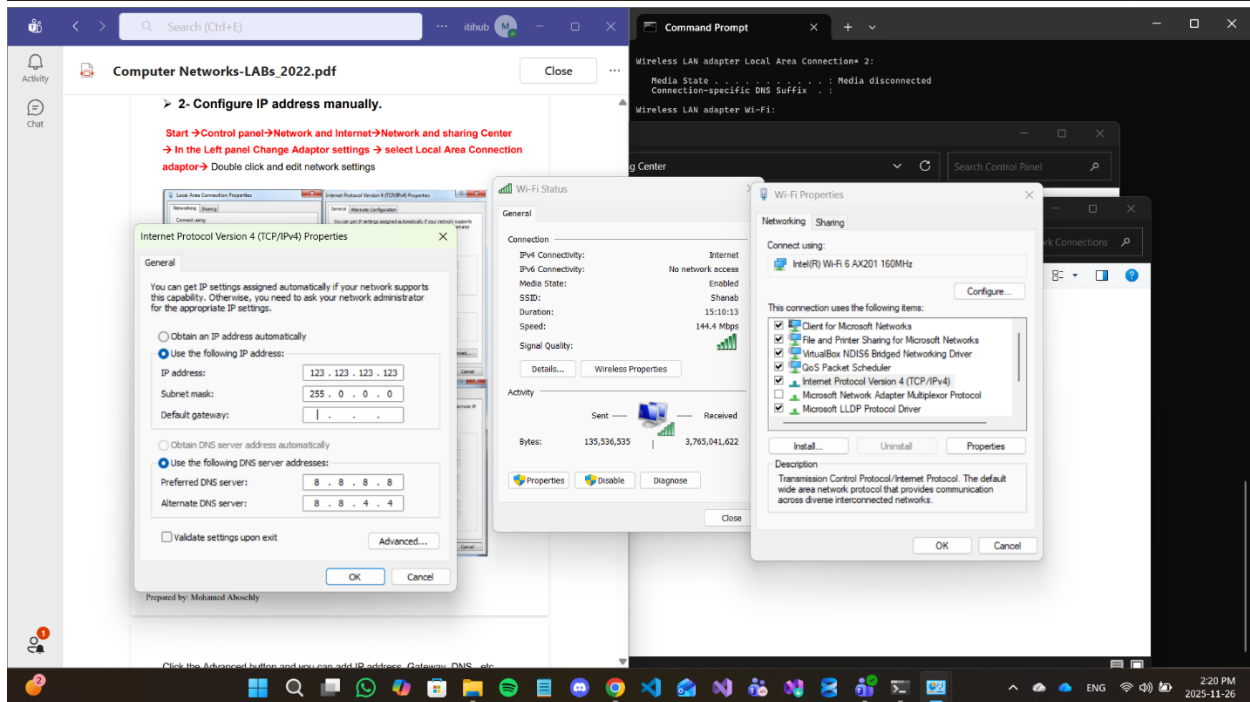
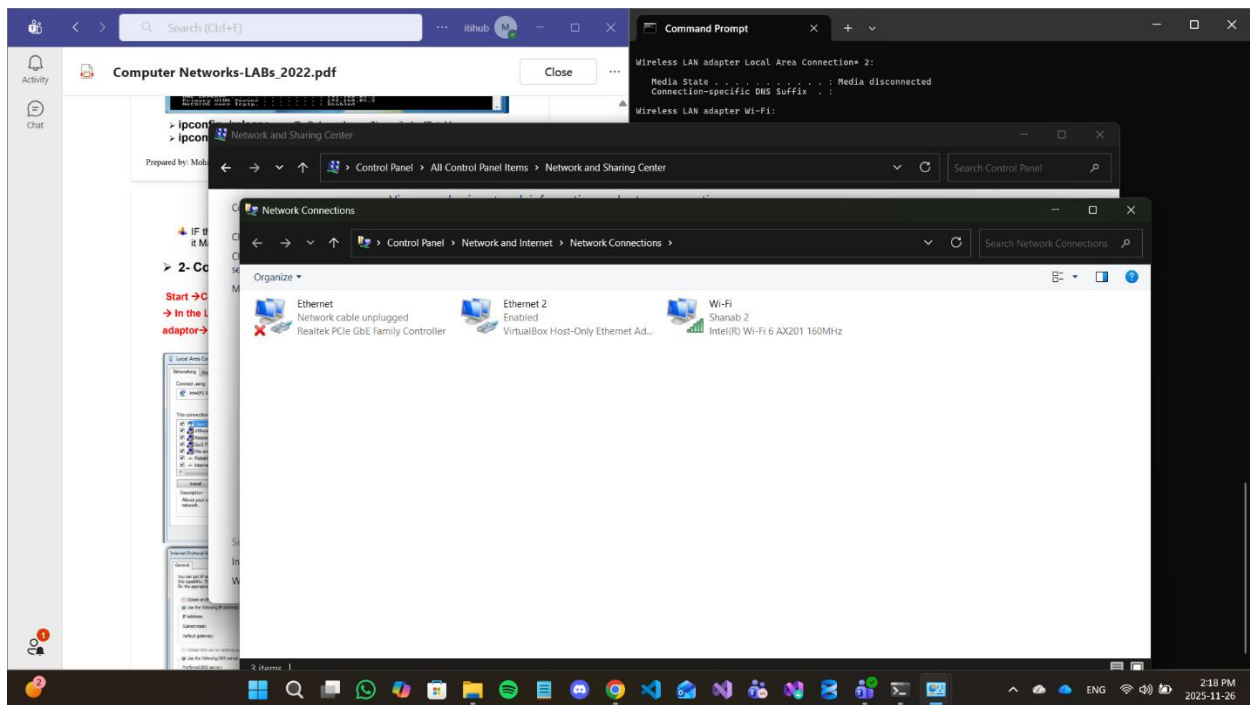
Connection-specific DNS Suffix . . : 
Link-local IPv6 Address . . . . . : fe80::e31:78bd:f1fe:15d%18
IPv6 Address. . . . . : 192.168.100.38
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.100.1

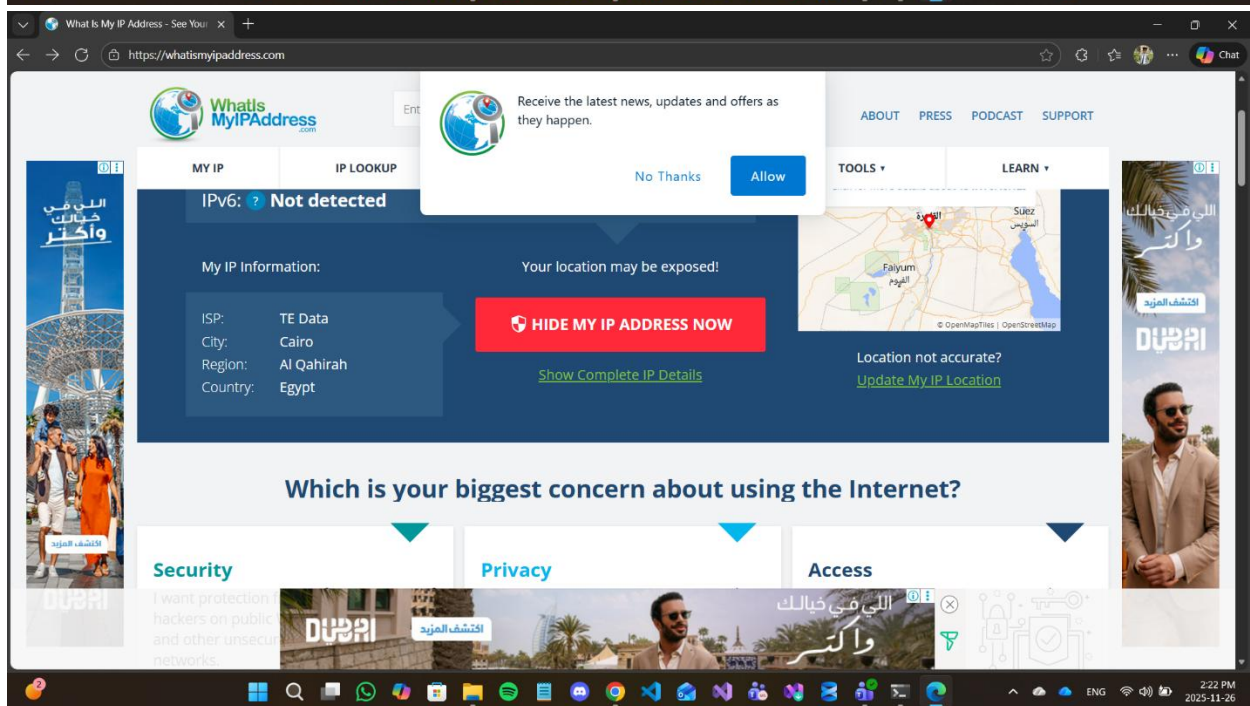
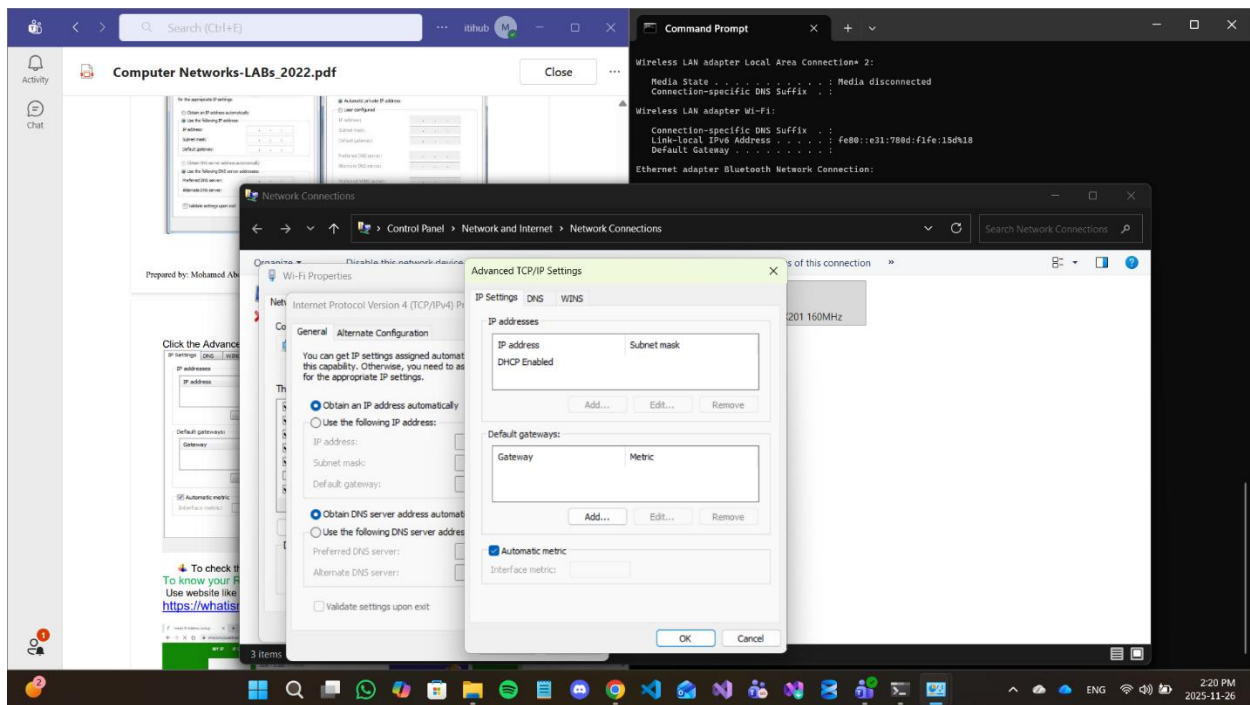
Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . . : 

C:\Users\mosta>
```

2:16 PM 2025-11-26





Computer Networks-LABs_2022.pdf

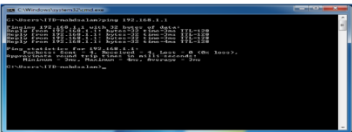
3- Ping Command

Check the connectivity

Ping Destination IP Of_Remote_Host

Suppose that the remote host has the ip 192.168.1.1

C:\>ping 192.168.1.1



Check the Availability of website

C:\>ping Yahoo.com

Continue ping operation unlimited (just add the switch "-t")

C:\>ping 192.168.1.1 -t

To exit write : ctrl+c

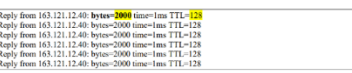
To control the number of ping packets, (just add the switch "-n" followed by the required packet number (space))

C:\>ping 192.168.1.1 -n 7

To control the size of ping packets, (just add the switch "-l" followed by the required packet size)

C:\>ping 163.121.25.40 -l 2000

C:\> Ping 163.121.12.40 -l 2000 -n 6



Prepared by: Mohamed Aboelsly

Command Prompt

C:\Users\mostar>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Request timed out.

Request timed out.

Request timed out.

Request timed out.

Ping statistics for 192.168.1.1:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\mostar>ping yahoo.com

Pinging yahoo.com [74.6.231.20] with 32 bytes of data:

Reply from 74.6.231.20: bytes=32 time=171ms TTL=51

Reply from 74.6.231.20: bytes=32 time=171ms TTL=51

Reply from 74.6.231.20: bytes=32 time=171ms TTL=51

Reply from 74.6.231.20: bytes=32 time=171ms TTL=51

Ping statistics for 74.6.231.20:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 171ms, Maximum = 171ms, Average = 171ms

C:\Users\mostar>ping yahoo.com -n 6

pinging yahoo.com [74.6.231.20] with 32 bytes of data:

Reply from 74.6.231.20: bytes=32 time=168ms TTL=51

Reply from 74.6.231.20: bytes=32 time=169ms TTL=51

Reply from 74.6.231.20: bytes=32 time=169ms TTL=51

Reply from 74.6.231.20: bytes=32 time=171ms TTL=51

Reply from 74.6.231.20: bytes=32 time=168ms TTL=51

Reply from 74.6.231.20: bytes=32 time=171ms TTL=51

Ping statistics for 74.6.231.20:

Packets: Sent = 6, Received = 6, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 168ms, Maximum = 171ms, Average = 169ms

C:\Users\mostar>ping yahoo.com -t

Pinging yahoo.com [74.6.231.20] with 32 bytes of data:

Reply from 74.6.231.20: bytes=32 time=168ms TTL=51

Reply from 74.6.231.20: bytes=32 time=169ms TTL=51

Reply from 74.6.231.20: bytes=32 time=169ms TTL=51

Reply from 74.6.231.20: bytes=32 time=171ms TTL=51

Reply from 74.6.231.20: bytes=32 time=168ms TTL=51

Reply from 74.6.231.20: bytes=32 time=171ms TTL=51

Ping statistics for 74.6.231.20:

Packets: Sent = 6, Received = 6, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 168ms, Maximum = 170ms, Average = 171ms

Ctrl+C

C

C:\Users\mostar>ping yahoo.com -l 2000

Pinging yahoo.com [74.6.231.20] with 2000 bytes of data:

Reply from 74.6.231.20: bytes=2000 time=172ms TTL=51

Reply from 74.6.231.20: bytes=2000 time=172ms TTL=51

Computer Networks-LABs_2022.pdf

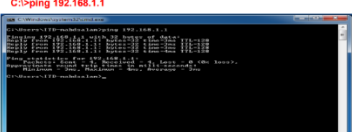
3- Ping Command

Check the connectivity

Ping Destination IP Of_Remote_Host

Suppose that the remote host has the ip 192.168.1.1

C:\>ping 192.168.1.1



Check the Availability of website

C:\>ping Yahoo.com

Continue ping operation unlimited (just add the switch "-t")

C:\>ping 192.168.1.1 -t

To exit write : ctrl+c

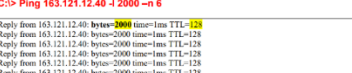
To control the number of ping packets, (just add the switch "-n" followed by the required packet number (space))

C:\>ping 192.168.1.1 -n 7

To control the size of ping packets, (just add the switch "-l" followed by the required packet size)

C:\>ping 163.121.25.40 -l 2000

C:\> Ping 163.121.12.40 -l 2000 -n 6



Prepared by: Mohamed Aboelsly

Command Prompt

C:\Users\mostar>ping yahoo.com -n 6

pinging yahoo.com [74.6.231.20] with 32 bytes of data:

Reply from 74.6.231.20: bytes=32 time=168ms TTL=51

Reply from 74.6.231.20: bytes=32 time=169ms TTL=51

Reply from 74.6.231.20: bytes=32 time=169ms TTL=51

Reply from 74.6.231.20: bytes=32 time=171ms TTL=51

Reply from 74.6.231.20: bytes=32 time=168ms TTL=51

Reply from 74.6.231.20: bytes=32 time=171ms TTL=51

Ping statistics for 74.6.231.20:

Packets: Sent = 6, Received = 6, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 168ms, Maximum = 170ms, Average = 169ms

Ctrl+C

C

C:\Users\mostar>ping yahoo.com -t

Pinging yahoo.com [74.6.231.20] with 32 bytes of data:

Reply from 74.6.231.20: bytes=32 time=168ms TTL=51

Reply from 74.6.231.20: bytes=32 time=169ms TTL=51

Reply from 74.6.231.20: bytes=32 time=169ms TTL=51

Reply from 74.6.231.20: bytes=32 time=171ms TTL=51

Reply from 74.6.231.20: bytes=32 time=168ms TTL=51

Reply from 74.6.231.20: bytes=32 time=171ms TTL=51

Ping statistics for 74.6.231.20:

Packets: Sent = 6, Received = 6, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 168ms, Maximum = 170ms, Average = 169ms

C:\Users\mostar>ping yahoo.com -l 2000

Pinging yahoo.com [74.6.231.20] with 2000 bytes of data:

Reply from 74.6.231.20: bytes=2000 time=172ms TTL=51

Reply from 74.6.231.20: bytes=2000 time=172ms TTL=51

Reply from 74.6.231.20: bytes=2000 time=172ms TTL=51

Reply from 74.6.231.20: bytes=2000 time=172ms TTL=51

Reply from 74.6.231.20: bytes=2000 time=172ms TTL=51

Reply from 74.6.231.20: bytes=2000 time=172ms TTL=51

Ping statistics for 74.6.231.20:

Packets: Sent = 6, Received = 6, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 170ms, Maximum = 175ms, Average = 171ms

C:\Users\mostar>ping yahoo.com -l 2000 -n 7

Pinging yahoo.com [74.6.231.20] with 2000 bytes of data:

Reply from 74.6.231.20: bytes=2000 time=171ms TTL=51

Reply from 74.6.231.20: bytes=2000 time=170ms TTL=51

Reply from 74.6.231.20: bytes=2000 time=170ms TTL=51

Reply from 74.6.231.20: bytes=2000 time=172ms TTL=51

Reply from 74.6.231.20: bytes=2000 time=172ms TTL=51

Reply from 74.6.231.20: bytes=2000 time=175ms TTL=51

Ping statistics for 74.6.231.20:

Packets: Sent = 7, Received = 7, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 170ms, Maximum = 175ms, Average = 171ms

C:\Users\mostar>

Computer Networks-LABs_2022.pdf

Activity

Chat

➤ To control the size of pinging packets, (just add the switch "-f" followed by the required packet size)

C:\>ping 163.121.25.40 -f 2000

C:\> Ping 163.121.12.40 -f 2000 -n 6

Reply from 163.121.12.40: bytes=2000 time=1ms TTL=128

Reply from 163.121.12.40: bytes=2000 time=1ms TTL=128

Reply from 163.121.12.40: bytes=2000 time=1ms TTL=128

Reply from 163.121.12.40: bytes=2000 time=1ms TTL=128

Reply from 163.121.12.40: bytes=2000 time=1ms TTL=128

Reply from 163.121.12.40: bytes=2000 time=1ms TTL=128

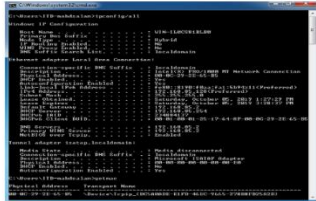
Prepared by: Mohamed Absobhy

➤ 4- MAC address

to know your current MAC (physical) address use the command:

➤ Ipconfig /all

➤ Get mac



➤ To know other devices MAC (physical) address use the ARP command:

➤ 5- ARP Command

Definition: ARP (Address Resolution Protocol) mapping an Internet Protocol (IP) address to its corresponding physical network address (Mac).

Command Prompt

Ethernet adapter Ethernet:

Media State Media disconnected

Connection-specific DNS Suffix

Description Realtek PCIe GBE Family Controller

Physical Address. 88-AA-C2-E3-35-76

DHCP Enabled. Yes

Autoconfiguration Enabled Yes

Ethernet adapter Ethernet 2:

Connection-specific DNS Suffix

Description VirtualBox Host-Only Ethernet Adapter

Physical Address. 0A-00-27-00-00-11

DHCP Enabled. No

Autoconfiguration Enabled Yes

Link-Local IPv6 Address fe80::f698:452c:cfe:1512v17(Preferred)

IPv6 Address. 192.168.56.1(Preferred)

Subnet Mask 255.255.255.0

Default Gateway

DHCPv6 IAID 755638119

DHCPv6 Client DUID. 00-01-00-01-38-1E-15-92-08-AA-C2-E3-SE-76

NetBIOS over Tcpip. Enabled

Wireless LAN adapter Local Area Connection* 1:

Media State Media disconnected

Connection-specific DNS Suffix

Description Microsoft Wi-Fi Direct Virtual Adapter

Physical Address. 8C-B8-7E-AE-85-EF

DHCP Enabled. Yes

Autoconfiguration Enabled Yes

Wireless LAN adapter Local Area Connection* 2:

Media State Media disconnected

Connection-specific DNS Suffix

Description Microsoft Wi-Fi Direct Virtual Adapter #2

Physical Address. 8C-B8-7E-AE-85-EF

DHCP Enabled. Yes

Autoconfiguration Enabled Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix

Description Intel(R) Wi-Fi 6 AX201 160MHz

Physical Address. 8C-B8-7E-AC-85-EE

DHCP Enabled. Yes

Autoconfiguration Enabled Yes

Link-Local IPv6 Address fe80::e31:788d:f1fe:15d1v18(Preferred)

IPv6 Address. 192.168.188.38(Preferred)

Subnet Mask 255.255.255.0

Lease Obtained. Wednesday, 26 November, 2025 2:16:21 PM

Lease Expires Thursday, 27 November, 2025 2:16:38 PM

Default Gateway 192.168.188.1

DHCP Server 192.168.188.1

DHCPv6 IAID 160217214

DHCPv6 Client DUID. 00-01-00-01-38-1E-15-92-08-AA-C2-E3-SE-76

DNS Servers 163.121.128.134

163.121.128.135

NetBIOS over Tcpip. Enabled

C:\Users\mosta\

2:25 PM

2025-11-26

Computer Networks-LABs_2022.pdf

Activity

Chat

➤ To control the size of pinging packets, (just add the switch "-f" followed by the required packet size)

C:\>ping 163.121.25.40 -f 2000

C:\> Ping 163.121.12.40 -f 2000 -n 6

Reply from 163.121.12.40: bytes=2000 time=1ms TTL=128

Reply from 163.121.12.40: bytes=2000 time=1ms TTL=128

Reply from 163.121.12.40: bytes=2000 time=1ms TTL=128

Reply from 163.121.12.40: bytes=2000 time=1ms TTL=128

Reply from 163.121.12.40: bytes=2000 time=1ms TTL=128

Reply from 163.121.12.40: bytes=2000 time=1ms TTL=128

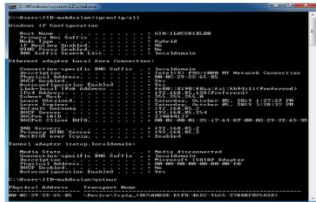
Prepared by: Mohamed Absobhy

➤ 4- MAC address

to know your current MAC (physical) address use the command:

➤ Ipconfig /all

➤ Get mac



➤ To know other devices MAC (physical) address use the ARP command:

➤ 5- ARP Command

Definition: ARP (Address Resolution Protocol) mapping an Internet Protocol (IP) address to its corresponding physical network address (Mac).

Command Prompt

Ethernet adapter Ethernet 2:

Connection-specific DNS Suffix VirtualBox Host-Only Ethernet Adapter

Description 8A-00-27-00-00-11

Physical Address. 8A-00-27-00-00-11

DHCP Enabled. No

Autoconfiguration Enabled Yes

Link-Local IPv6 Address fe80::f698:452c:cfe:1512v17(Preferred)

IPv6 Address. 192.168.56.1(Preferred)

Subnet Mask 255.255.255.0

Default Gateway

DHCPv6 IAID 755638119

DHCPv6 Client DUID. 00-01-00-01-38-1E-15-92-08-AA-C2-E3-SE-76

NetBIOS over Tcpip. Enabled

Wireless LAN adapter Local Area Connection* 1:

Media State Media disconnected

Connection-specific DNS Suffix

Description Microsoft Wi-Fi Direct Virtual Adapter

Physical Address. 8C-B8-7E-AC-85-E7

DHCP Enabled. Yes

Autoconfiguration Enabled Yes

Wireless LAN adapter Local Area Connection* 2:

Media State Media disconnected

Connection-specific DNS Suffix

Description Microsoft Wi-Fi Direct Virtual Adapter #2

Physical Address. 8C-B8-7E-AC-85-E7

DHCP Enabled. Yes

Autoconfiguration Enabled Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix

Description Intel(R) Wi-Fi 6 AX201 160MHz

Physical Address. 8C-B8-7E-AC-85-EE

DHCP Enabled. Yes

Autoconfiguration Enabled Yes

Link-Local IPv6 Address fe80::e31:788d:f1fe:15d1v18(Preferred)

IPv6 Address. 192.168.188.38(Preferred)

Subnet Mask 255.255.255.0

Lease Obtained. Wednesday, 26 November, 2025 2:16:21 PM

Lease Expires Thursday, 27 November, 2025 2:16:38 PM

Default Gateway 192.168.188.1

DHCP Server 192.168.188.1

DHCPv6 IAID 160217214

DHCPv6 Client DUID. 00-01-00-01-38-1E-15-92-08-AA-C2-E3-SE-76

DNS Servers 163.121.128.134

163.121.128.135

NetBIOS over Tcpip. Enabled

C:\Users\mosta>getmac

Physical Address Transport Name

=====

8C-B8-7E-AC-85-E7 \Device\Tcpip_{F8241CC3-8DC2-4F38-B7CA-C611B93A02E8}

8B-AA-C2-E3-35-76 Media disconnected

8A-00-27-00-00-11 \Device\Tcpip_{0575DFCD-F2E1-45D6-A568-6229944681B}

C:\Users\mosta\

2:25 PM

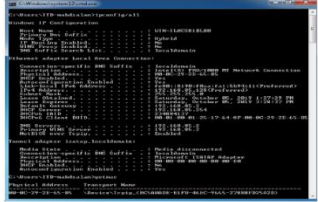
2025-11-26

Computer Networks-LABs_2022.pdf

4- MAC address

to know your current MAC (physical) address use the command:

- ↳ `ipconfig /all`
- ↳ `Get mac`



To know other devices MAC (physical) address use the **ARP** command:

5- ARP Command

Definition: ARP (Address Resolution Protocol) mapping an Internet Protocol (IP) address to its corresponding physical network address (MAC)

ARP request is broadcast, an ARP reply is unicast

Run the following command to view the contents of the ARP cache (run as Admin)

`C:\>arp -a`

C. To delete the arp cache write the command

`C:\>arp -d`

Command Prompt

```
Link-Local IPv6 Address . . . . . fe80::f6b0:4b2c:cfe:1512%17(Preferred)
IPv6 Address. . . . . 192.168.56.1(Preferred)
Subnet Mask . . . . . 255.255.255.0
Default Gateway . . . . .
DHCPv6 ID . . . . . 755036119
DHCPv6 Client DUID . . . . . 08-01-80-01-38-1E-15-92-88-A4-C2-E3-5E-76
NetBIOS over Tcpip. . . . . Enabled

Wireless LAN adapter Local Area Connection* 1:
Media State . . . . . Media disconnected
Connection-specific DNS Suffix . . . .
Description . . . . . Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . 8C-B8-7E-AC-85-EF
DHCP Enabled. . . . . Yes
Autoconfiguration Enabled . . . . . Yes

Wireless LAN adapter Local Area Connection* 2:
Media State . . . . . Media disconnected
Connection-specific DNS Suffix . . . .
Description . . . . . Microsoft Wi-Fi Direct Virtual Adapter #2
Physical Address. . . . . 8C-B8-7E-AC-85-EF
DHCP Enabled. . . . . Yes
Autoconfiguration Enabled . . . . . Yes

Wireless LAN adapter Wi-Fi:
Connection-specific DNS Suffix . . . .
Description . . . . . Intel(R) Wi-Fi 6 AX201 160MHz
Physical Address. . . . . 8C-B8-7E-AC-85-E2
Link-Local IPv6 Address . . . . . fe80::c31:788d:f1fe:15d%18(Preferred)
IPv6 Address. . . . . 192.168.188.38(Preferred)
Subnet Mask . . . . . 255.255.255.0
Default Gateway . . . . . 192.168.188.1
Lease Expires . . . . . Thursday, 27 November, 2025 2:16:21 PM
Lease Obtained . . . . . Thursday, 27 November, 2025 2:16:38 PM
DHCPv6 ID . . . . . 1460217214
DHCPv6 Client DUID . . . . . 08-01-80-01-38-1E-15-92-88-A4-C2-E3-5E-76
DHCP Server . . . . . 192.168.188.1
DNS Servers . . . . . 163.121.128.134
NetBIOS over Tcpip. . . . . Enabled

C:\Users\mostafa>arp -a

Physical Address Transport Name
=====
8C-B8-7E-AC-85-E2 \Device\NPF{F82A1CC3-8DC2-4F38-8FCA-C611B93A2E8}
8B-A4-C2-E3-55-76 \Device\NPF{0575DFCD-F2E1-45D6-A568-6229944E681B}
8A-08-27-08-00-11 \Device\NPF{0575DFCD-F2E1-45D6-A568-6229944E681B}

C:\Users\mostafa>arp -v

Connection Name Network Adapter Physical Address Transport Name
=====
Wi-Fi Intel(R) Wi-Fi 6 AX201 160MHz 8C-B8-7E-AC-85-E2 \Device\NPF{F82A1CC3-8DC2-4F38-8FCA-C611B93A2E8}
Ethernet Realtek PCIe Gb 8B-A4-C2-E3-55-76 Media disconnected
Ethernet 2 VirtualBox Host 8A-08-27-08-00-11 \Device\NPF{0575DFCD-F2E1-45D6-A568-6229944E681B}

C:\Users\mostafa>
```

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5- ARP Command

Definition: ARP (Address Resolution Protocol) mapping an Internet Protocol (IP) address to its corresponding physical network address (MAC)

ARP request is broadcast, an ARP reply is unicast

Run the following command to view the contents of the ARP cache (run as Admin)

`C:\>arp -a`

C. To delete the arp cache write the command

`C:\>arp -d`

Command Prompt

```
Interface: 192.168.56.1 --- 0x11
Internet Address Physical Address Type
192.168.56.255 ff-ff-ff-ff-ff-ff static
224.0.0.22 01-00-5e-00-00-16 static
224.0.0.251 01-00-5e-00-00-fb static
224.0.0.252 01-00-5e-00-00-fc static
239.255.255.258 01-00-5e-7f-ff-fa static

Interface: 192.168.188.38 --- 0x12
Internet Address Physical Address Type
192.168.188.1 08-00-1c-1b-c5-ad Dynamic
192.168.188.255 ff-ff-ff-ff-ff-ff static
224.0.0.22 01-00-5e-00-00-16 static
224.0.0.251 01-00-5e-00-00-fb static
224.0.0.252 01-00-5e-00-00-fc static
239.255.255.258 01-00-5e-7f-ff-fa static
255.255.255.255 ff-ff-ff-ff-ff-ff static

C:\Users\mostafa>arp -d
The ARP entry deletion failed: The requested operation requires elevation.

C:\Users\mostafa>
```

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To know other devices MAC (physical) address use the **ARP** command:

5- ARP Command

Definition: ARP (Address Resolution Protocol) mapping an Internet Protocol (IP) address to its corresponding physical network address (Mac)

ARP request is **broadcast**, an ARP reply is **unicast**

Run the following command to view the contents of the ARP cache **run as Admin**

```
C:\>arp -a
```

C. To delete the arp cache write the command

```
C:\>arp -d
```

Prepared by: Mohamed Aboeshly

6- Routing Table Commands

A routing table is a set of rules, often viewed in table format, that is used to determine where data packets traveling over an Internet Protocol (IP) network will be directed. All IP-enabled devices, including routers and switches, use routing tables.

Command Prompt

```
C:\Users\mosta>arp -a
```

Interface: 192.168.56.1 --- 0x11		
Internet Address	Physical Address	Type
192.168.56.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.258	01-00-5e-7f-ff-fa	static

```
C:\Users\mosta>arp -d
```

The ARP entry deletion failed: The requested operation requires elevation.

```
C:\Users\mosta>arp -a
```

Interface: 192.168.56.1 --- 0x11		
Internet Address	Physical Address	Type
192.168.56.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
239.255.255.258	01-00-5e-7f-ff-fa	static

Interface: 192.168.100.30 --- 0x12		
Internet Address	Physical Address	Type
192.168.100.1	04-8c-16-1b-c6-ad	dynamic
192.168.100.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.258	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

C:\Users\mosta>|

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Administrator: Command Prompt

```
Microsoft Windows [Version 10.0.26200.7171]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>arp -d

C:\Windows\System32>|
```

Prepared by: Mohamed Aboeshly

6- Routing Table Commands

A routing table is a set of rules, often viewed in table format, that is used to determine where data packets traveling over an Internet Protocol (IP) network will be directed. All IP-enabled devices, including routers and switches, use routing tables.

Command Prompt

```
C:\Users\mosta>arp -a
```

Interface: 192.168.56.1 --- 0x11		
Internet Address	Physical Address	Type
192.168.56.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
239.255.255.258	01-00-5e-7f-ff-fa	static

Interface: 192.168.100.30 --- 0x12		
Internet Address	Physical Address	Type
192.168.100.1	04-8c-16-1b-c6-ad	dynamic
192.168.100.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.258	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

The requested operation requires elevation.

Interface: 192.168.56.1 --- 0x11		
Internet Address	Physical Address	Type
192.168.56.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
239.255.255.258	01-00-5e-7f-ff-fa	static

Interface: 192.168.100.30 --- 0x12		
Internet Address	Physical Address	Type
192.168.100.1	04-8c-16-1b-c6-ad	dynamic
192.168.100.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.258	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

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Internet Adapter Name: eth0

IP Address: 192.168.100.1

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.100.30

MAC Address: 08:00:27:00:00:11

Internet Adapter Name: eth1

IP Address: 192.168.100.1

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.100.30

MAC Address: 08:00:27:00:00:11

To know other devices MAC (physical) address use the **ARP** command:

5- ARP Command

Definition: **ARP (Address Resolution Protocol)** mapping an Internet Protocol (IP) address to its corresponding physical network address (Mac)

ARP request is broadcast; an ARP reply is **unicast**

Run the following command to view the contents of the ARP cache **run as Admin**

C:\>arp -a

C. To delete the arp cache write the command

C:\>arp -d

Prepared by: Mohamed Aboelsly

6- Routing Table Commands

A routing table is a set of rules, often viewed in table format, that is used to determine where data packets traveling over an Internet Protocol (IP) network will be directed. All IP-enabled devices, including routers and switches, use routing tables.

Command Prompt

C:\Users\mosta>arp -a

Interface: 192.168.56.1 --- 0x11

Internet Address Physical Address Type

192.168.56.255 ff-ff-ff-ff-ff-ff static

224.0.0.22 01-00-5e-00-00-16 static

224.0.0.251 01-00-5e-00-00-00 static

224.0.0.252 01-00-5e-00-00-fc static

239.255.255.258 01-00-5e-7f-ff-fa static

Interface: 192.168.100.30 --- 0x12

Internet Address Physical Address Type

192.168.100.1 04-8c-16-1b-c6-ad dynamic

192.168.100.255 ff-ff-ff-ff-ff-ff static

224.0.0.22 01-00-5e-00-00-16 static

224.0.0.251 01-00-5e-00-00-fb static

224.0.0.252 01-00-5e-00-00-fc static

239.255.255.258 01-00-5e-7f-ff-fa static

255.255.255.255 ff-ff-ff-ff-ff-ff static

C:\Users\mosta>arp -d

The ARP entry deletion failed: The requested operation requires elevation.

C:\Users\mosta>arp -a

Interface: 192.168.56.1 --- 0x11

Internet Address Physical Address Type

192.168.56.255 ff-ff-ff-ff-ff-ff static

224.0.0.22 01-00-5e-00-00-16 static

239.255.255.258 01-00-5e-7f-ff-fa static

Interface: 192.168.100.30 --- 0x12

Internet Address Physical Address Type

192.168.100.1 04-8c-16-1b-c6-ad dynamic

192.168.100.255 ff-ff-ff-ff-ff-ff static

224.0.0.22 01-00-5e-00-00-16 static

239.255.255.258 01-00-5e-7f-ff-fa static

C:\Users\mosta>arp -a

Interface: 192.168.56.1 --- 0x11

Internet Address Physical Address Type

192.168.56.255 ff-ff-ff-ff-ff-ff static

224.0.0.22 01-00-5e-00-00-16 static

Interface: 192.168.100.30 --- 0x12

Internet Address Physical Address Type

192.168.100.1 04-8c-16-1b-c6-ad dynamic

192.168.100.255 ff-ff-ff-ff-ff-ff static

224.0.0.22 01-00-5e-00-00-16 static

C:\Users\mosta>]

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C:\>arp -d

Prepared by: Mohamed Aboelsly

6- Routing Table Commands

A routing table is a set of rules, often viewed in table format, that is used to determine where data packets traveling over an Internet Protocol (IP) network will be directed. All IP-enabled devices, including routers and switches, use routing tables.

- To view the local routing table, write the command

C:\>route print

- To add a route to the routing table, write the command

Destination Net ID mask IP GW IP metric value, IP # "Interface # or IP

Destination Net ID mask IP GW IP metric value

C:\>Route add 163.121.12 mask 255.255.255.0 163.121.12.1 metric 1 (Most trusted)

C:\>Route add 163.121.12 mask 255.255.255.0 163.121.12.1 metric 10 (Not trusted)

Problem: With route add command: Element Not found - parameter is in correct Solving:

Adding new route using Net ID NOT Host ID

Route add 163.121.12 mask 255.255.255.0 163.121.12.1 >>> **fail**

Route add 163.121.12 mask 255.255.255.0 163.121.12.1 >>> **Wrong**

163.121.12.10 to Host IP not Host ID

Metric is switch can be added to the end of the add commands to differentiate between the routes to the same destinations. You can assign this value from 0 to 255 the default value is 20. The lowest value is most trusted.

- To add a default route, write the command

C:\>route add 0.0.0.0 mask 0.0.0.0 163.121.25.1

- To Show the default route before adding new route:

C:\>Route print 0.0.0.0

Command Prompt

C:\Users\mosta>route print

=====

Interface List

10...88 a4 c2 e3 5e 76Realtek PCIe GbE Family Controller

17...4a 00 27 00 00 11VirtualBox Host-Only Ethernet Adapter

14...8c b8 7e ae 85 efMicrosoft Wi-Fi Direct Virtual Adapter #1

10...8e b8 7e ae 85 eeMicrosoft Wi-Fi Direct Virtual Adapter #2

16...8c b8 7e ae 85 eeIntel(R) Wi-Fi 6 AX201 160MHz

1.....Software Loopback Interface 1

=====

IPv4 Route Table

=====

Active Routes:

Network Destination Netmask Gateway Interface Metric

0.0.0.0 0.0.0.0 192.168.100.1 192.168.100.30 50

127.0.0.0 255.0.0.0 On-Link 127.0.0.1 331

127.0.0.1 255.255.255.255 On-Link 127.0.0.1 331

127.255.255.255 255.255.255.255 On-Link 127.0.0.1 331

192.168.56.0 255.255.255.0 On-Link 192.168.56.1 281

192.168.56.1 255.255.255.255 On-Link 192.168.56.1 281

192.168.100.0 255.255.255.0 On-Link 192.168.100.30 386

192.168.100.30 255.255.255.255 On-Link 192.168.100.30 386

192.168.100.255 255.255.255.255 On-Link 192.168.100.30 386

224.0.0.0 240.0.0.0 On-Link 127.0.0.1 331

224.0.0.0 240.0.0.0 On-Link 192.168.100.30 386

255.255.255.255 255.255.255.255 On-Link 127.0.0.1 331

255.255.255.255 255.255.255.255 On-Link 192.168.56.1 281

255.255.255.255 255.255.255.255 On-Link 192.168.100.30 386

255.255.255.255 255.255.255.255 On-Link 192.168.100.30 386

=====

Persistent Routes:

None

=====

IPv6 Route Table

=====

Active Routes:

If Metric Network Destination Gateway

1 331 ::1/128 On-Link

17 281 fe80::f6b8:452c:cfe:1512/128 On-Link

18 386 fe80::f6b8:452c:cfe:1512/128 On-Link

18 386 fe80::f6b8:452c:cfe:1512/128 On-Link

17 281 fe80::f6b8:452c:cfe:1512/128 On-Link

1 331 ff98::/8 On-Link

17 281 ff98::/8 On-Link

18 386 ff98::/8 On-Link

=====

Persistent Routes:

None

C:\Users\mosta>]

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6- Routing Table Commands

A routing table is a set of rules, often viewed in table format, that is used to determine where data packets traveling over an Internet Protocol (IP) network will be directed. All IP-enabled devices, including routers and switches, use routing tables.

- To view the local routing table, write the command
C:\>route print
- To add a route to the routing table, write the command
Destination Net ID mask IP GW IP metric value, IP # "Interface IP" or IP IP
Destination Net ID mask IP GW IP metric value
C:\>Route add 163.121.12 mask 255.255.255.0 163.121.12.1 metric 10 (Not tested)
- C:\>Route add 163.121.12 mask 255.255.255.0 163.121.12.1 metric 10** (Not tested)

Problem: With route add command: Element Not found -- parameter is in correct

Solving:

Adding new route using Net ID NOT Host ID

Route add 163.121.12 mask 255.255.255.0 163.121.12.1 >>> **OK!**

Route add 163.121.12 mask 255.255.255.0 163.121.12.1 >>> **Wrong**

163.121.12.1 is Host IP not Net ID

Metric: is switch can be added to the end of the add commands to differentiate between the routes to the same destinations. You can assign this value from 0 to 255 the default value is 20. The lowest value is most trusted.

- To add a default route, write the command
C:\>route add 0.0.0.0 mask 0.0.0.0 192.168.1.1
- To Show the default route before adding new route:
C:\>Route print 0.0.0.0

Add new default route: will not delete the old default route but add another one with another metric

- To delete a certain route, write the command
C:\>route delete 157.0.0.0
- To Delete default route:
C:\> route delete 0.0.0.0 → **we delete all default routes**

Prepared by: Mohamed Aboachly

Command Prompt

```
127.255.255.255 255.255.255.255 On-Link 127.0.0.1 331
192.168.56.0 255.255.255.0 On-Link 192.168.56.1 281
192.168.56.1 255.255.255.0 On-Link 192.168.56.1 281
192.168.56.255 255.255.255.255 On-Link 192.168.56.1 281
192.168.160.0 255.255.255.0 On-Link 192.168.160.30 386
192.168.160.30 255.255.255.255 On-Link 192.168.160.30 386
192.168.160.255 255.255.255.255 On-Link 192.168.160.30 386
224.0.0.0 240.0.0.0 On-Link 127.0.0.1 331
224.0.0.8 240.0.0.0 On-Link 192.168.56.1 281
224.0.0.16 240.0.0.0 On-Link 192.168.160.30 386
255.255.255.255 255.255.255.255 On-Link 127.0.0.1 331
255.255.255.255 255.255.255.255 On-Link 192.168.56.1 281
255.255.255.255 255.255.255.255 On-Link 192.168.160.30 386
=====
Persistent Routes:
None

IPv6 Route Table
=====
Active Routes:
Interface Metric Network Destination Gateway
1 331 ::1/128 On-Link
17 281 f6b0::f64 On-Link
18 386 f6b0::f64 On-Link
18 386 f6b0::e31:78bd:f1fe:15d/128 On-Link
17 281 f6b0::f6b0:452:c1fe:1512/128 On-Link
1 331 f6b0::f6 On-Link
17 281 f6b0::f6 On-Link
18 386 f6b0::f6 On-Link
=====
Persistent Routes:
None

C:\Users\mosta>route print 10.10.10.0
=====
Interface List
10 {80 a4 c2 e3 5e 76} Realtek PCIe GbE Family Controller
17 {da 00 27 00 00 11} VirtualBox Host-Only Ethernet Adapter
18 {8c b8 7e ae 85 ef} Microsoft Wi-Fi Direct Virtual Adapter #2
19 {8c b8 7e ae 85 ee} Microsoft Wi-Fi Direct Virtual Adapter #2
16 {8c b8 7e ae 85 ee} Intel(R) Wi-Fi 6 AX201 160MHz
1 {80 00 00 00 00 00} Software Loopback Interface 1
=====
IPv4 Route Table
=====
Active Routes:
Network Destination Netmask Gateway Interface Metric
10.10.10.0 255.255.255.0 192.168.1.1 192.168.160.30 51
=====
Persistent Routes:
None

IPv6 Route Table
=====
Active Routes:
None
Persistent Routes:
None

C:\Users\mosta>
```

Administrator: Command Prompt

```
Microsoft Windows [Version 10.0.26200.7171]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>route add 10.10.10.0 mask 255.255.255.0 192.168.1.1
OK!

C:\Windows\System32>route delete 10.10.10.0
OK!

C:\Windows\System32>
```

Command Prompt

```
55 On-Link 192.168.56.1 281
55 On-Link 192.168.160.30 386
55 On-Link 192.168.160.30 386
55 On-Link 192.168.160.30 386
55 On-Link 127.0.0.1 331
55 On-Link 192.168.56.1 281
55 On-Link 192.168.160.30 386
55 On-Link 127.0.0.1 331
55 On-Link 192.168.56.1 281
55 On-Link 192.168.160.30 386
=====
Gateway
On-Link
On-Link
5d/128
On-Link
512/128
On-Link
On-Link
On-Link
On-Link
=====
10.0
=====
Realtek PCIe GbE Family Controller
VirtualBox Host-Only Ethernet Adapter
Microsoft Wi-Fi Direct Virtual Adapter
Microsoft Wi-Fi Direct Virtual Adapter #2
Intel(R) Wi-Fi 6 AX201 160MHz
Software Loopback Interface 1
=====
IPv4 Route Table
=====
Active Routes:
Network Destination Netmask Gateway Interface Metric
10.10.10.0 255.255.255.0 192.168.1.1 192.168.160.30 51
=====
Persistent Routes:
None

IPv6 Route Table
=====
Active Routes:
None
Persistent Routes:
None

C:\Users\mosta>route delete 10.10.10.0
The requested operation requires elevation.

C:\Users\mosta>
```

Prepared by: Mohamed Aboachly

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6- Routing Table Commands

A routing table is a set of rules, often viewed in table format, that is used to determine where data packets traveling over an Internet Protocol (IP) network will be directed. All IP-enabled devices, including routers and switches, use routing tables.

- To view the local routing table, write the command
C:\>route print
- To add a route to the routing table, write the command
Destination Net ID mask IP GW IP metric value, IP # "Interface I" or IF IP
Destination Net ID mask IP GW IP metric value
C:\>Route add 163.121.12 mask 255.255.255.0 163.121.12.1 metric 1 (Not correct)
C:\>Route add 163.121.12 mask 255.255.255.0 163.121.12.1 >>> 10 (Not correct)

Solving:
Problem: With route add command: Element Not found -- parameter is in correct

Adding new route using Net ID NOT Host ID
Route add 163.121.12 mask 255.255.255.0 163.121.12.1 >>> 10
Route add 163.121.12 mask 255.255.255.0 163.121.12.1 >>> 10
163.121.12.10 is Host IP not Net ID

Metric: is switch can be added to the end of the add commands to differentiate between the routes to the same destinations. You can assign this value from 0 to 255 the default value is 20. The lowest value is most trusted.

- To add a default route, write the command
C:\>route add 0.0.0.0 mask 0.0.0.0 163.121.25.1
- To Show the default route before adding new route:
C:\>Route print 0.0.0.0

Add new default route: will not delete the old default route but add another one with another metric

- To delete a certain route, write the command
C:\>route delete 157.0.0.0
- To Delete default route:
C:\> route delete 0.0.0.0 >>> will delete all default routes

Prepared by: Mohamed Abochely

Command Prompt

```
None

IPv6 Route Table
=====
Active Routes:
None
Persistent Routes:
None

C:\Users\mosta>route print
=====
Interface List
10...88 a4 c2 e3 5e 76 .....Realtek PCIe GbE Family Controller
17...0a 00 27 00 00 11 .....VirtualBox Host-Only Ethernet Adapter
14...8c b8 7e ae 85 ef .....Microsoft Wi-Fi Direct Virtual Adapter #2
19...8e b8 7e ae 85 ee .....Microsoft Wi-Fi Direct Virtual Adapter #2
18...8c b8 7e ae 85 ee .....Intel(R) Wi-Fi 6 AX201 160MHz
.....Software Loopback Interface 1
=====

IPv4 Route Table
=====
Active Routes:
Network Destination        Netmask          Gateway          Interface        Metric
0.0.0.0                    0.0.0.0          192.168.100.1    192.168.100.30    50
127.0.0.0                  255.0.0.0        On-Link          127.0.0.1         331
127.0.0.1                  255.0.0.0        On-Link          127.0.0.1         331
127.0.0.0                  255.0.0.0        On-Link          127.0.0.1         331
127.255.255.255            255.255.255.0    On-Link          192.168.100.30    386
192.168.56.1               255.255.255.0    On-Link          192.168.56.1      281
192.168.56.1               255.255.255.0    On-Link          192.168.56.1      281
192.168.100.0              255.255.255.0    On-Link          192.168.100.30    386
192.168.100.30             255.255.255.0    On-Link          192.168.100.30    386
224.0.0.0                  240.0.0.0        On-Link          127.0.0.1         331
224.0.0.0                  240.0.0.0        On-Link          192.168.56.1      281
224.0.0.0                  240.0.0.0        On-Link          192.168.100.30    386
255.255.255.255            255.255.255.0    On-Link          127.0.0.1         331
255.255.255.255            255.255.255.0    On-Link          192.168.56.1      281
255.255.255.255            255.255.255.0    On-Link          192.168.100.30    386

Persistent Routes:
None

IPv6 Route Table
=====
Active Routes:
If Metric Network Destination      Gateway
1 331 ::1 ::1
17 281 f68b::/64 On-Link
20 386 f68b::/64 On-Link
18 386 f68b::x31:788d::f1fe:15d/128 On-Link

17 281 f68b::f68b:4b2c:cfe:1512/128 On-Link
1 331 f68b::/8 On-Link
17 281 f68b::/8 On-Link
18 386 f68b::/8 On-Link

Persistent Routes:
None

C:\Users\mosta>
```

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- To Delete default routes:
C:\> route delete 0.0.0.0 >>> will delete all default routes

Prepared by: Mohamed Abochely

Command Prompt

```
C:\Users\mosta>netstat -n

Active Connections
Proto Local Address          Foreign Address         State
TCP    192.168.100.30:40455    52.98.200.146:443      ESTABLISHED
TCP    192.168.100.30:40604    52.123.135.184:443      ESTABLISHED
TCP    192.168.100.30:40915    52.123.135.28:443       ESTABLISHED
TCP    192.168.100.30:53044    52.182.143.215:443      TIME_WAIT
TCP    192.168.100.30:53192    52.112.100.54:443       ESTABLISHED
TCP    192.168.100.30:54125    52.112.122.33:443       ESTABLISHED
TCP    192.168.100.30:54126    52.112.122.39:443       ESTABLISHED
TCP    192.168.100.30:54129    52.123.243.16:443       ESTABLISHED
TCP    192.168.100.30:55193    97.153.125.174:443      ESTABLISHED
TCP    192.168.100.30:55194    52.123.200.64:443       ESTABLISHED
TCP    192.168.100.30:55196    52.112.122.39:443       ESTABLISHED
TCP    192.168.100.30:55208    98.66.133.186:443       ESTABLISHED
TCP    192.168.100.30:55209    98.66.133.186:443       ESTABLISHED
TCP    192.168.100.30:55220    98.66.133.186:443       ESTABLISHED
TCP    192.168.100.30:56041    52.98.200.226:443       ESTABLISHED
TCP    192.168.100.30:59507    52.123.137.9:443        ESTABLISHED
TCP    192.168.100.30:59693    9.141.5.27:443          ESTABLISHED
TCP    192.168.100.30:61866    52.98.163.69:443        ESTABLISHED
TCP    192.168.100.30:61887    13.69.116.107:443       ESTABLISHED
TCP    192.168.100.30:61889    28.42.65.85:443         ESTABLISHED
TCP    192.168.100.30:63832    52.123.243.18:443       ESTABLISHED
TCP    192.168.100.30:63259    108.177.15.188:5228     ESTABLISHED
TCP    192.168.100.30:64216    52.123.129.14:443       ESTABLISHED
TCP    192.168.100.30:64444    156.200.37.34:443       CLOSE_WAIT
TCP    192.168.100.30:64445    156.200.37.98:443       CLOSE_WAIT
TCP    192.168.100.30:64446    182.172.97.54:443       CLOSE_WAIT
TCP    192.168.100.30:64447    156.200.39.98:443       CLOSE_WAIT
TCP    192.168.100.30:64448    156.200.35.163:443      CLOSE_WAIT
TCP    192.168.100.30:64449    156.200.35.228:443      CLOSE_WAIT
TCP    192.168.100.30:64450    156.200.39.34:443       CLOSE_WAIT
TCP    192.168.100.30:64451    182.172.103.68:443      CLOSE_WAIT
TCP    192.168.100.30:64452    197.240.193.56:443      CLOSE_WAIT
TCP    192.168.100.30:64453    182.172.97.54:443       CLOSE_WAIT
TCP    192.168.100.30:64458    52.123.128.14:443       ESTABLISHED
TCP    192.168.100.30:65294    52.97.168.210:443       ESTABLISHED

C:\Users\mosta>
```

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But:
C:\> route delete 0.0.0.0 mask 0.0.0.0 163.121.25.1 >>> will delete default route only

Prepared by: Mohamed Abochely

7- Opened ports and sessions

netstat (network statistics) is a command-line tool that displays network connections (both incoming and outgoing), routing tables, and a number of network interface statistics. It is available on Unix, Unix-like, and Windows NT-based operating systems.

To know about the concurrent TCP connections on my PC, write the following command
C:\>netstat -n
Displays addresses and port numbers in numerical form.

To know about the concurrent TCP connections on my PC, write the following command
C:\>netstat -n
C:\>netstat -n IP address >>> C:\>netstat -n 192.168.1.1

8- Port Scanning

You can use any port scanner to scan ports locally or on a remote PC, try using the IP Scanner.

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To Delete default route:
C:\> route delete 0.0.0.0 → will delete all default routes

Prepared by: Mohamed Aloschly

But:
C:\> route delete 0.0.0.0 mask 0.0.0.0 163.121.25.1 → will delete default route only

7- Opened ports and sessions

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To know about the concurrent TCP connections on my PC, write the following command
C:\> netstat -n

Displays addresses and port numbers in numerical form.

To know about the concurrent TCP connections on my PC, write the following command
C:\> netstat -n

C:\> netstat -a ip address → C:\> netstat -a 192.168.1.1

Microsoft Windows [Version 6.0.6002]
Copyright (c) 2004 Microsoft Corporation. All rights reserved.

C:\> netstat -n

Active Connections

Proto Local Address Foreign Address State

TCP 192.168.1.1:14411 192.168.1.1:14411 CLOSE_WAIT

TCP 192.168.1.1:14412 192.168.1.1:14412 CLOSE_WAIT

8- Port Scanning

You can use any port scanner to scan ports locally or on a remote PC, try using the IP Scanner.

Command Prompt - netstat -n

Active Connections

Proto Local Address Foreign Address State

TCP 192.168.100.30:49455 52.98.200.146:443 ESTABLISHED

TCP 192.168.100.30:49494 52.123.135.184:443 ESTABLISHED

TCP 192.168.100.30:49915 52.123.135.28:443 ESTABLISHED

TCP 192.168.100.30:53844 52.182.143.215:443 TIME_WAIT

TCP 192.168.100.30:53192 52.112.108.54:443 ESTABLISHED

TCP 192.168.100.30:54125 52.112.122.33:443 ESTABLISHED

TCP 192.168.100.30:54126 52.112.122.39:443 ESTABLISHED

TCP 192.168.100.30:54129 52.123.243.16:443 ESTABLISHED

TCP 192.168.100.30:55193 57.153.125.174:443 ESTABLISHED

TCP 192.168.100.30:55194 52.123.208.64:443 ESTABLISHED

TCP 192.168.100.30:55196 52.112.122.39:443 ESTABLISHED

TCP 192.168.100.30:55209 98.66.133.186:443 ESTABLISHED

TCP 192.168.100.30:55209 98.66.133.186:443 ESTABLISHED

TCP 192.168.100.30:55220 98.66.133.186:443 ESTABLISHED

TCP 192.168.100.30:56461 52.98.200.226:443 ESTABLISHED

TCP 192.168.100.30:59557 52.123.137.9:443 ESTABLISHED

TCP 192.168.100.30:59693 9.141.5.27:443 ESTABLISHED

TCP 192.168.100.30:61886 52.94.103.59:443 ESTABLISHED

TCP 192.168.100.30:61887 13.69.116.107:443 ESTABLISHED

TCP 192.168.100.30:61889 20.42.65.85:443 ESTABLISHED

TCP 192.168.100.30:63832 52.123.243.18:443 ESTABLISHED

TCP 192.168.100.30:63359 108.177.15.188:5228 ESTABLISHED

TCP 192.168.100.30:64216 52.123.129.14:443 ESTABLISHED

TCP 192.168.100.30:64404 156.208.37.34:443 CLOSE_WAIT

TCP 192.168.100.30:64405 156.208.37.98:443 CLOSE_WAIT

TCP 192.168.100.30:64406 102.132.97.54:443 CLOSE_WAIT

TCP 192.168.100.30:64407 156.208.39.98:443 CLOSE_WAIT

TCP 192.168.100.30:64408 156.208.35.163:443 CLOSE_WAIT

TCP 192.168.100.30:64409 156.208.35.228:443 CLOSE_WAIT

TCP 192.168.100.30:64451 182.132.97.54:443 CLOSE_WAIT

TCP 192.168.100.30:64452 157.248.195.56:443 CLOSE_WAIT

TCP 192.168.100.30:64453 182.132.97.54:443 CLOSE_WAIT

TCP 192.168.100.30:65858 52.123.128.14:443 ESTABLISHED

TCP 192.168.100.30:65294 52.97.168.218:443 ESTABLISHED

C:\Users\mosta>netstat -a

Active Connections

Proto Local Address Foreign Address State

TCP 0.0.0.0:135 Shanab:0 LISTENING

TCP 0.0.0.0:445 Shanab:0 LISTENING

TCP 0.0.0.0:5840 Shanab:0 LISTENING

TCP 0.0.0.0:7680 Shanab:0 LISTENING

TCP 0.0.0.0:49464 Shanab:0 LISTENING

TCP 0.0.0.0:49465 Shanab:0 LISTENING

TCP 0.0.0.0:49467 Shanab:0 LISTENING

TCP 0.0.0.0:49468 Shanab:0 LISTENING

TCP 0.0.0.0:49478 Shanab:0 LISTENING

TCP 127.0.0.1:9222 Shanab:0 LISTENING

TCP 192.168.100.30:139 Shanab:0 LISTENING

TCP 192.168.100.30:139 Shanab:0 LISTENING

TCP 192.168.100.30:49455 52.98.200.146:https ESTABLISHED

TCP 192.168.100.30:49494 52.123.135.184:https ESTABLISHED

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To Delete default route:
C:\> route delete 0.0.0.0 → will delete all default routes

Prepared by: Mohamed Aloschly

But:
C:\> route delete 0.0.0.0 mask 0.0.0.0 163.121.25.1 → will delete default route only

7- Opened ports and sessions

netstat (network statistics) is a command-line tool that displays network connections (both incoming and outgoing), routing tables, and a number of network interface statistics. It is available on Unix, Unix-like, and Windows NT-based operating systems.

To know about the concurrent TCP connections on my PC, write the following command
C:\> netstat -n

Displays addresses and port numbers in numerical form.

To know about the concurrent TCP connections on my PC, write the following command
C:\> netstat -n

C:\> netstat -a ip address → C:\> netstat -a 192.168.1.1

Microsoft Windows [Version 6.0.6002]
Copyright (c) 2004 Microsoft Corporation. All rights reserved.

C:\> netstat -n

Active Connections

Proto Local Address Foreign Address State

TCP 192.168.1.1:14411 192.168.1.1:14411 CLOSE_WAIT

TCP 192.168.1.1:14412 192.168.1.1:14412 CLOSE_WAIT

8- Port Scanning

You can use any port scanner to scan ports locally or on a remote PC, try using the IP Scanner.

Command Prompt

Proto Local Address Foreign Address State

TCP 192.168.100.30:49455 52.98.200.146:443 ESTABLISHED

TCP 192.168.100.30:49494 52.123.135.184:443 ESTABLISHED

TCP 192.168.100.30:49915 52.123.135.28:443 ESTABLISHED

TCP 192.168.100.30:53844 52.182.143.215:443 TIME_WAIT

TCP 192.168.100.30:53192 52.112.108.54:443 ESTABLISHED

TCP 192.168.100.30:54125 52.112.122.33:443 ESTABLISHED

TCP 192.168.100.30:54129 52.123.243.16:443 ESTABLISHED

TCP 192.168.100.30:55193 57.153.125.174:443 ESTABLISHED

TCP 192.168.100.30:55194 52.123.208.64:443 ESTABLISHED

TCP 192.168.100.30:55196 52.112.122.39:443 ESTABLISHED

TCP 192.168.100.30:55209 98.66.133.186:443 ESTABLISHED

TCP 192.168.100.30:55220 98.66.133.186:443 ESTABLISHED

TCP 192.168.100.30:56461 52.98.200.226:443 ESTABLISHED

TCP 192.168.100.30:59557 52.123.137.9:443 ESTABLISHED

TCP 192.168.100.30:59693 9.141.5.27:443 ESTABLISHED

TCP 192.168.100.30:61886 52.94.103.59:443 ESTABLISHED

TCP 192.168.100.30:61887 13.69.116.107:443 ESTABLISHED

TCP 192.168.100.30:61889 20.42.65.85:443 ESTABLISHED

TCP 192.168.100.30:63832 52.123.243.18:443 ESTABLISHED

TCP 192.168.100.30:63359 108.177.15.188:5228 ESTABLISHED

TCP 192.168.100.30:64216 52.123.129.14:443 ESTABLISHED

TCP 192.168.100.30:64404 156.208.37.34:443 CLOSE_WAIT

TCP 192.168.100.30:64405 156.208.37.98:443 CLOSE_WAIT

TCP 192.168.100.30:64406 102.132.97.54:443 CLOSE_WAIT

TCP 192.168.100.30:64407 156.208.39.98:443 CLOSE_WAIT

TCP 192.168.100.30:64408 156.208.35.163:443 CLOSE_WAIT

TCP 192.168.100.30:64409 156.208.35.228:443 CLOSE_WAIT

TCP 192.168.100.30:64451 182.132.97.54:443 CLOSE_WAIT

TCP 192.168.100.30:64452 157.248.195.56:443 CLOSE_WAIT

TCP 192.168.100.30:64453 182.132.97.54:443 CLOSE_WAIT

TCP 192.168.100.30:65858 52.123.128.14:443 ESTABLISHED

TCP 192.168.100.30:65294 52.97.168.218:443 ESTABLISHED

C:\Users\mosta>netstat -a

Active Connections

Proto Local Address Foreign Address State

TCP 0.0.0.0:135 Shanab:0 LISTENING

TCP 0.0.0.0:445 Shanab:0 LISTENING

TCP 0.0.0.0:5840 Shanab:0 LISTENING

TCP 0.0.0.0:7680 Shanab:0 LISTENING

TCP 0.0.0.0:49464 Shanab:0 LISTENING

TCP 0.0.0.0:49465 Shanab:0 LISTENING

TCP 0.0.0.0:49467 Shanab:0 LISTENING

TCP 0.0.0.0:49468 Shanab:0 LISTENING

TCP 0.0.0.0:49478 Shanab:0 LISTENING

TCP 127.0.0.1:9222 Shanab:0 LISTENING

TCP 192.168.100.30:139 Shanab:0 LISTENING

TCP 192.168.100.30:139 Shanab:0 LISTENING

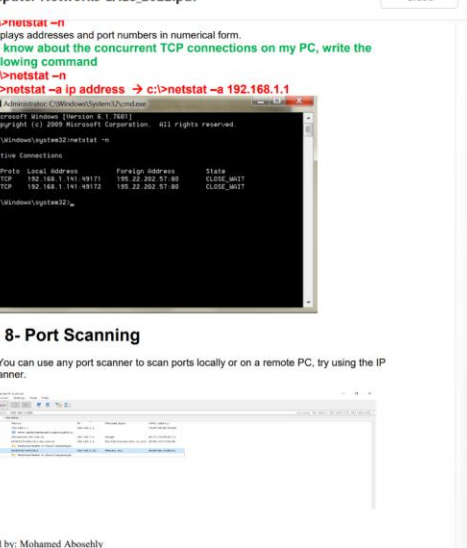
TCP 192.168.100.30:49455 52.98.200.146:https ESTABLISHED

TCP 192.168.100.30:49494 52.123.135.184:https ESTABLISHED

TCP 192.168.100.30:53844 52.182.143.215:https TIME_WAIT

TCP 192.168.100.30:53192 52.112.108.54:https ESTABLISHED

TCP 192.168.100.30:54125 52.112.122.33:https ESTABLISHED



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c:\netstat -n
Display addresses and port numbers in numerical form.
To know about the common TCP connections on my PC, write the following command
c:\netstat -n
c:\netstat -n → ip address → **c:\netstat -a 192.168.1.1**

```

C:\Users\Ahsan>netstat -n

Active Connections

Proto Local Address          Foreign Address        State
TCP    192.168.1.1:80           192.168.1.1:80         LISTENING
TCP    192.168.1.1:443         192.168.1.1:443        LISTENING
TCP    192.168.1.1:1433        192.168.1.1:1433       LISTENING
TCP    192.168.1.1:1434        192.168.1.1:1434       LISTENING
TCP    192.168.1.1:1435        192.168.1.1:1435       LISTENING
TCP    192.168.1.1:1436        192.168.1.1:1436       LISTENING
TCP    192.168.1.1:1437        192.168.1.1:1437       LISTENING
TCP    192.168.1.1:1438        192.168.1.1:1438       LISTENING
TCP    192.168.1.1:1439        192.168.1.1:1439       LISTENING
TCP    192.168.1.1:1440        192.168.1.1:1440       LISTENING
TCP    192.168.1.1:1441        192.168.1.1:1441       LISTENING
TCP    192.168.1.1:1442        192.168.1.1:1442       LISTENING
TCP    192.168.1.1:1443        192.168.1.1:1443       LISTENING
TCP    192.168.1.1:1444        192.168.1.1:1444       LISTENING
TCP    192.168.1.1:1445        192.168.1.1:1445       LISTENING
TCP    192.168.1.1:1446        192.168.1.1:1446       LISTENING
TCP    192.168.1.1:1447        192.168.1.1:1447       LISTENING
TCP    192.168.1.1:1448        192.168.1.1:1448       LISTENING
TCP    192.168.1.1:1449        192.168.1.1:1449       LISTENING
TCP    192.168.1.1:1450        192.168.1.1:1450       LISTENING
TCP    192.168.1.1:1451        192.168.1.1:1451       LISTENING
TCP    192.168.1.1:1452        192.168.1.1:1452       LISTENING
TCP    192.168.1.1:1453        192.168.1.1:1453       LISTENING
TCP    192.168.1.1:1454        192.168.1.1:1454       LISTENING
TCP    192.168.1.1:1455        192.168.1.1:1455       LISTENING
TCP    192.168.1.1:1456        192.168.1.1:1456       LISTENING
TCP    192.168.1.1:1457        192.168.1.1:1457       LISTENING
TCP    192.168.1.1:1458        192.168.1.1:1458       LISTENING
TCP    192.168.1.1:1459        192.168.1.1:1459       LISTENING
TCP    192.168.1.1:1460        192.168.1.1:1460       LISTENING
TCP    192.168.1.1:1461        192.168.1.1:1461       LISTENING
TCP    192.168.1.1:1462        192.168.1.1:1462       LISTENING
TCP    192.168.1.1:1463        192.168.1.1:1463       LISTENING
TCP    192.168.1.1:1464        192.168.1.1:1464       LISTENING
TCP    192.168.1.1:1465        192.168.1.1:1465       LISTENING
TCP    192.168.1.1:1466        192.168.1.1:1466       LISTENING
TCP    192.168.1.1:1467        192.168.1.1:1467       LISTENING
TCP    192.168.1.1:1468        192.168.1.1:1468       LISTENING
TCP    192.168.1.1:1469        192.168.1.1:1469       LISTENING
TCP    192.168.1.1:1470        192.168.1.1:1470       LISTENING
TCP    192.168.1.1:1471        192.168.1.1:1471       LISTENING
TCP    192.168.1.1:1472        192.168.1.1:1472       LISTENING
TCP    192.168.1.1:1473        192.168.1.1:1473       LISTENING
TCP    192.168.1.1:1474        192.168.1.1:1474       LISTENING
TCP    192.168.1.1:1475        192.168.1.1:1475       LISTENING
TCP    192.168.1.1:1476        192.168.1.1:1476       LISTENING
TCP    192.168.1.1:1477        192.168.1.1:1477       LISTENING
TCP    192.168.1.1:1478        192.168.1.1:1478       LISTENING
TCP    192.168.1.1:1479        192.168.1.1:1479       LISTENING
TCP    192.168.1.1:1480        192.168.1.1:1480       LISTENING
TCP    192.168.1.1:1481        192.168.1.1:1481       LISTENING
TCP    192.168.1.1:1482        192.168.1.1:1482       LISTENING
TCP    192.168.1.1:1483        192.168.1.1:1483       LISTENING
TCP    192.168.1.1:1484        192.168.1.1:1484       LISTENING
TCP    192.168.1.1:1485        192.168.1.1:1485       LISTENING
TCP    192.168.1.1:1486        192.168.1.1:1486       LISTENING
TCP    192.168.1.1:1487        192.168.1.1:1487       LISTENING
TCP    192.168.1.1:1488        192.168.1.1:1488       LISTENING
TCP    192.168.1.1:1489        192.168.1.1:1489       LISTENING
TCP    192.168.1.1:1490        192.168.1.1:1490       LISTENING
TCP    192.168.1.1:1491        192.168.1.1:1491       LISTENING
TCP    192.168.1.1:1492        192.168.1.1:1492       LISTENING
TCP    192.168.1.1:1493        192.168.1.1:1493       LISTENING
TCP    192.168.1.1:1494        192.168.1.1:1494       LISTENING
TCP    192.168.1.1:1495        192.168.1.1:1495       LISTENING
TCP    192.168.1.1:1496        192.168.1.1:1496       LISTENING
TCP    192.168.1.1:1497        192.168.1.1:1497       LISTENING
TCP    192.168.1.1:1498        192.168.1.1:1498       LISTENING
TCP    192.168.1.1:1499        192.168.1.1:1499       LISTENING
TCP    192.168.1.1:1500        192.168.1.1:1500       LISTENING
TCP    192.168.1.1:1501        192.168.1.1:1501       LISTENING
TCP    192.168.1.1:1502        192.168.1.1:1502       LISTENING
TCP    192.168.1.1:1503        192.168.1.1:1503       LISTENING
TCP    192.168.1.1:1504        192.168.1.1:1504       LISTENING
TCP    192.168.1.1:1505        192.168.1.1:1505       LISTENING
TCP    192.168.1.1:1506        192.168.1.1:1506       LISTENING
TCP    192.168.1.1:1507        192.168.1.1:1507       LISTENING
TCP    192.168.1.1:1508        192.168.1.1:1508       LISTENING
TCP    192.168.1.1:1509        192.168.1.1:1509       LISTENING
TCP    192.168.1.1:1510        192.168.1.1:1510       LISTENING
TCP    192.168.1.1:1511        192.168.1.1:1511       LISTENING
TCP    192.168.1.1:1512        192.168.1.1:1512       LISTENING
TCP    192.168.1.1:1513        192.168.1.1:1513       LISTENING
TCP    192.168.1.1:1514        192.168.1.1:1514       LISTENING
TCP    192.168.1.1:1515        192.168.1.1:1515       LISTENING
TCP    192.168.1.1:1516        192.168.
```